

ABHITAY SHINDE

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EDUCATION

Columbia University

M.S. Data Science, GPA: 3.7/4.0

- Relevant Courses: Agentic AI, Fintech and Data Economy (PhD elective), Big Data Analytics, and Applied Machine Learning.

Sep 2024 - Dec 2025

New York, NY

PROFESSIONAL EXPERIENCE

Julius Baer

Data Scientist (Generative AI and Personalization)

New York, NY

Jun 2025 - Sep 2025

- Engineered an LLM personalization chatbot using LangGraph and prompt chaining with retrieval-augmented context, automating content curation pipelines for relationship managers and cutting selection time by 35%.
- Uncovered a 25% lift in engagement by running causal and exploratory analysis on customer behavior with Python and SQL.
- Standardized guardrails, and reproducibility across cross-functional teams by establishing an experimentation framework.

TD Bank

Data Scientist Capstone (Customer Analytics and Segmentation)

New York, NY

Jan 2025 - May 2025

- Validated fraud-detection strategies by executing causal inference and A/B testing on 100M+ transactions in PySpark.
- Elevated fraud model accuracy from 0.20 to 0.67 PR-AUC using an XGBoost model with SHAP-based explainability.
- Informed senior leadership on segmentation, money laundering, and fraud-reduction strategy through data-driven insights.

Ask2AI

Data Scientist (Customer Acquisition)

New York, NY

Feb 2026 - June 2026

- Built a feature selection and causal inference pipeline in Python and Polars correcting selection bias in the bank's default model, integrating 11 data sources across 30M+ customers.
- Inferred default probability for rejected customers using Double Machine Learning with propensity-corrected features, validated at 0.86 AUC on a held-out cohort.

Quantitative Research Co-op (Customer Acquisition)

Jan 2025 - May 2025

- Improved ROI attribution across acquisition channels by 22% by deploying uplift modeling for personalization and difference-in-differences to isolate causal impact, identifying optimal credit limit thresholds to lift conversion rates.
- Boosted customer LTV by 12% and reduced churn by 25% by building uplift models and MILP for acquisition allocation.

E-Revbay Pvt. Ltd.

Founding IT Intern (Acquisition and Retention Analytics)

Mumbai, IN

Dec 2021 - May 2022

- Increased qualified customer leads by 50% by leading acquisition analytics pipelines using Equifax data, SQL, and Python.
- Eliminated 25% of manager intervention and accelerated response by building real-time root-cause pipelines in Tableau.

SELECTED PROJECTS

IBM Agentic Library - Graph RAG for Document Q&A | LangGraph, RAG, Neo4j, FAISS, ChromaDB, LLMs, Python, OpenAI

- Architected Graph RAG pipeline for IBM's open-source agentic library using Neo4j and FAISS for document retrieval.
- Modeled documents as graphs capturing entity relationships, improving retrieval precision 30%+ over naive vector search.
- Orchestrated agentic pipelines using LangGraph for multi-step reasoning, decomposed tasks via CoT prompting.

Causal Product Analytics Suite | Python, Causal Inference, A/B Testing, Propensity Score Matching, DiD, Regression, MILP

- Built an MMM estimating incremental LTV across channels, optimizing revealed different ranking vs. last-touch attribution.
- Prevented an incorrect ship by using PSM-based DiD to a change showing 110% lift, post-matching retention revealed 0%.
- Averted an incorrect rollback by running analysis on a 15.4% decline, isolating cohort and seasonality as the primary drivers.

LEADERSHIP EXPERIENCE

- Winner of global AI hackathon, placing 1st out of 750 teams with international recognition for applied AI.
- Founding team member for analytics products (MyLoanBhai, TestMyPolicy, FinQy 2.0), scaling across 35 cities, 3 countries.
- Lead author on 4+ peer-reviewed IEEE papers with 15+ citations, spanning predictive maintenance and ML modeling.

TECHNICAL SKILLS

- Generative AI & LLMs:** RAG, LangChain, LangGraph, Agentic AI, Prompt Engineering, Hugging Face, ChromaDB.
- Machine Learning:** Regression, XGBoost, Random Forest, GNN, Clustering, PCA, Forecasting, A/B Testing, SHAP.
- Data Engineering:** Python, SQL, PySpark, Pandas, NumPy, Airflow, Tableau, Docker, Git, Jupyter, Databricks.
- Analytics & Experimentation:** Causal Analysis, Root Cause Analysis, Metric Design, Hypothesis Testing, PSM, DiD.
- Cloud & Infrastructure:** AWS, GCP, Vertex AI, BigQuery, MLOps, Data Pipelines, Databricks.